THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS TWO-CHAPTER TEST SET C

Chapters 1 & 2: **Orienting Yourself: The Use of Coordinates, Introduction to Linear Polynomials** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The point $(8,\ 1)$ lies: 1

(a) Quadrant II **(b)** Quadrant III **(c)** Quadrant IV **(d)** Quadrant I

2. The distance between $A(-2,\ 3)$ and $B(3,\ -9)$ is: 1

(a) 15 units **(b)** 12 units **(c)** 13 units **(d)** 14 units

3. The distance between $A(2,\ -1)$ and $B(10,\ 14)$ is: 1

(a) 17 units **(b)** 18 units **(c)** 16 units **(d)** 19 units

4. The midpoint of the segment joining $A(0,\ -8)$ and $B(6,\ 0)$ is: 1

(a) $(3,\ 4)$ **(b)** $(4,\ -3)$ **(c)** $(6,\ -8)$ **(d)** $(3,\ -4)$

5. $M(-1,\ 4)$ is the midpoint of AB and $A(5,\ 6)$. The coordinates of B are: 1

(a) $(-7,\ -2)$ **(b)** $(-6,\ -2)$ **(c)** $(4,\ 10)$ **(d)** $(-7,\ 2)$

6. The degree of the polynomial $4z - 3$ is: 1

(a) 1 **(b)** 0 **(c)** 2 **(d)** 3

7. The value of the linear polynomial $3x + 1$ at $x = 4$ is: 1

(a) 16 **(b)** 11 **(c)** -13 **(d)** 13

8. The value of $7x^2 - 4x + 6$ at $x = 0$ is: 1

(a) 8 **(b)** 3 **(c)** 11 **(d)** 6

9. The sequence $3, 7, 11, 15, \ldots$ forms a linear pattern. Its nth term is: 1

(a) $3n + 4$ **(b)** $4n - 7$ **(c)** $4n - 1$ **(d)** $4n + 3$

10. The sequence $7, 12, 17, 22, \ldots$ forms a linear pattern. Its nth term is: 1

(a) $5n + 2$ **(b)** $5n + 7$ **(c)** $7n + 5$ **(d)** $5n - 12$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Is the point $(-5,\ -12)$ at the same distance from the origin as $(13,\ 0)$? Justify. 2

12. The points $A(2,\ 0)$, $B(0,\ 2)$ and $C(-2,\ 0)$ are joined to form a triangle. Name the type of triangle with reason. 2

13. Show that $(7,\ 15)$ lies on the line $y = 2x + 1$ but $(3,\ 8)$ does not. 2

14. Write the number of tiles in stage n if the pattern $1, 3, 5, 7, \ldots$ is followed, and find which stage has 47 tiles. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A school ground is plotted on a grid with the flag-post at $F(0, 0)$, the main gate at $G(8, 0)$ and the library at $L(0, 6)$. Find the distance between the gate and the library and the midpoint of the gate–library path. 3

16. The volume of a box of length 7 cm and breadth 11 cm is a linear polynomial in the height h. Find the volume for $h = 5, 9, 13$. 3

17. Find the perimeter of the triangle with vertices $(0,\ 0)$, $(4,\ 0)$ and $(4,\ 3)$. 3

OR

Three vertices of a rectangle are $(2,\ 1)$, $(8,\ 1)$ and $(8,\ 5)$. Find the fourth vertex and the length of the diagonal. 3

18. Sarita has a 500-page book and reads 20 pages a day. Write the pages left after d days and find the days needed to finish. 3

OR

A student has ₹500 in her account and receives ₹150 every month. Write a linear expression for the amount in the nth month (from the second month onwards) and find in which month she will have ₹1,850. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The vertices of $\triangle ABC$ are $A(-2,\ 0)$, $B(2,\ 0)$ and $C(0,\ 2)$, and those of $\triangle PQR$ are $P(-4, 0)$, $Q(4, 0)$, $R(0, 4)$. Show that $\triangle ABC \sim \triangle PQR$ by finding the ratios of the sides. 5

OR

Find the value(s) of k if the distance between $(k,\ 2)$ and $(3,\ -4)$ is 10 units, and write the corresponding points. 5

20. (i) Find the value of $5x^2 - 3x + 7$ at $x = 1$ and $x = -2$. 5
(ii) Write a cubic polynomial in x whose x^2 coefficient is -7.
(iii) State the degree of $4z^3 + 5z^2 - 11$ and the coefficient of z.

OR

Case study: A student has ₹500 in her savings account and gets ₹150 every month. 5
(i) Write a linear expression for the amount after n months.
(ii) Find the amount after 6 months.
(iii) After how many months will she have ₹2,000?
(iv) Plot (describe) the graph and identify the slope.

END OF QUESTION PAPER — **Total: 40 Marks**